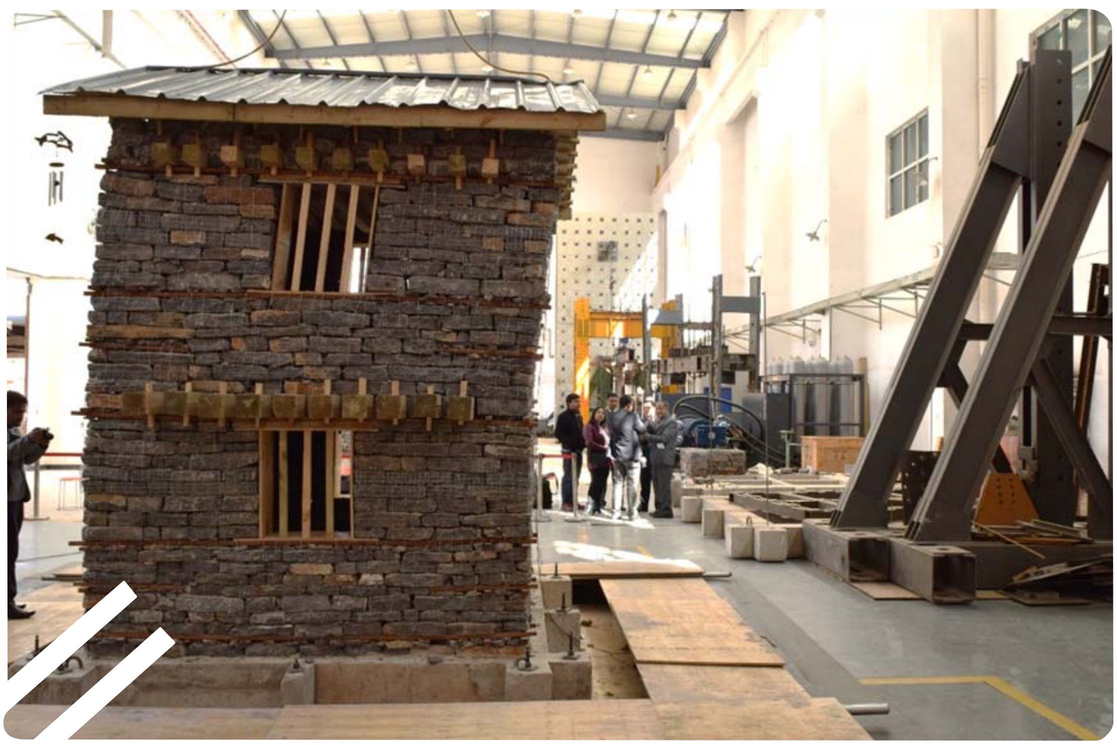
Draft Report on Collaborative Project of ICCR-DRR between NSET and BNU



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1 INTRODUCTION

National Society for Earthquake Technology-Nepal (NSET) is jointly working with Beijing Normal University, China in a collaborative research project of International Center for Collaborative Research on Disaster Risk Reduction ICCR-DRR on the project titled "Development, testing, demonstration and training of better-built procedures and retrofit techniques for non-engineered housing in urban and rural areas of the Himalayan belt" with the financial support from the UK's Department for International Development DFID. The project commenced on July 2016 and is scheduled to complete on September 2017. The study is on low strength stone masonry buildings which got extensive damage in Gorkha Earthquake 2015. The main purpose of the study is to do laboratory experimental tests on stone masonry buildings to understand the behavior of the buildings in earthquake and help develop a proper evidence based guideline for the construction of such buildings.

2 PROJECT BACKGROUND AND RATIONAL OF THE STUDY

Amongst the various building types, stone masonry buildings are the most vulnerable proved from past earthquake experiences in Nepal and the region (Pakistan earthquake 2005, Taplejung earthquake Sept 2011 and Gorkha earthquake April 2015 etc). Hence a proper know how on this type of building to resist strong earthquake is inevitable when there is a concern for re-construction of hundreds of thousands of such buildings after 2015 Gorkha earthquake.

Earthquake resistant techniques for stone masonry buildings are given in Nepal National Building Code NBC 203:1994 (Originally developed) and NBC 203:2015 (Recently revised) in the form of "Guidelines for earthquake resistant building construction: Low Strength Masonry". The guideline is for the new building construction and mud based. It does not talk about dry stone masonry.

The research project is on shake table test on stone masonry buildings using timber as earthquake resistant element. The rationale of the study is described below.

Stone masonry building structures are inherently vulnerable to earthquake shaking resulting in brittle failure as seen in past earthquakes (Pakistan earthquake 2005 Taplejung earthquake Sept 2011 and Gorkha earthquake April 2015). Majority of casualty in 2015 Gorkha earthquake is due to collapse of stone masonry buildings, some mountainous regions of Nepal almost 100% of stone masonry buildings collapsed. One example is Sindhupalchowk district where 100% such buildings collapsed resulting in 3423 deaths which is 40% of total deaths. Similar is the situation in other mountainous districts such as Nuwakot, Dhading, Rasuwa, Gorkha, Kavrepalanchok etc. where many stone masonry buildings failed causing numerous injuries and deaths Lessons of the 2015 Nepal Earthquake Disaster, short report by Sujan Malla). Similarly, 95% of the total collapsed houses and 67.7% of the damaged houses in Gorkha earthquake are low strength masonry buildings (Table 1.4: Building Typologies and Damage, Vol. 2, PDNA report; National Planning Commission 2015).

If the local people start building their houses in the traditional manner, it will again get damaged in future earthquake. Hence build back safer is very important in these region where stone masonry is the only option due to economic consideration being locally available.

1. To minimize the losses of lives and property in rural areas during inevitable future earthquakes it is very important to start constructing earthquake resistant buildings. Nepal National Building code (NBC) was originally developed long back in 1994 with the incorporation of earthquake resistant elements in this type of buildings. The code is revised recently in 2015 but is mostly based on conceptual background and hypothetical judgment and no tests (experimental or past earthquake) are done in real sense. Before

implementation, it requires a study: Detail analytical study to be followed by lab experiments to substantiate the study outcome from computer analysis.

2. Nepal National Building code has the provision for masonry buildings for its seismic safety the use of vertical bars at corners and jambs of door/window openings for in-plane loading. There was also the concern on using the vertical bars at corners within the wall section in low strength stone masonry as this is likely to affect the integrity of the adjoining walls rather than improving. Mud mortar and no mortar in case of dry stone do not provide bonding between stones. The use of vertical reinforcement at the center within the wall section is also practically difficult during construction. The alternate option is to use vertical bars on the sides of wall and connection as a replacement of vertical reinforcement bar within the wall section at the center. This can prove to be more effective than the option of using vertical bar at the center of the wall. But this idea needs verification by experimental tests.
3. There has been extensive studies and researches on Reinforced Concrete Cement frame building structures worldwide but very little study on masonry buildings and literally negligible on stone buildings. NSET is involved in analytical studies to understand the behavior of such stone masonry buildings at different intensities of shaking. This needs verification at experimental level. The shake table test is targeted to have a full understanding of the behavior of such buildings with earthquake resistant elements.

3 PROJECT OBJECTIVES

The main objective of the project is to understand the behavior of low strength stone masonry buildings by experimental and analytical studies using different techniques as proposed in Nepal National Building Code and with improvised options to ensure Life Safety of the building occupants. The specific objectives are:

- An accurate review of construction methods in the regions, highlighting construction details that lead to failures as well as the ones that prove effective in limiting damage to housings;
- A literature review of strengthening methods for vernacular construction, taking examples and lessons learned in other parts of the world.
- Development of prototypes of strengthening techniques and new construction details, to be tested to prove their reliability.
- Site demonstration on how to do it and learn from onsite testing
- Development of training course and training materials for construction industry and self-built capacity building.

4 COMPLETED TASKS

The following tasks have been completed by the end of January 2017.

1. Review of Existing Construction Method, Building Codes and Practices in Nepal and Experiences of China

   The construction practice before Gorkha Earthquake 2015 and reconstruction of stone masonry buildings after the earthquake was studied in rural areas of Nepal and mainly in the earthquake affected areas. Depending on the site and location of the building, the practice is to use both mud based random rubble and regular flat stones or dry stone buildings. Earthquake resistant elements such as horizontal and vertical bands were hardly used in stone masonry buildings before Gorkha earthquake, hence the building code was not followed in rural areas. After Gorkha earthquake, many governmental and non-governmental organizations conducted mason trainings and various other capacity building trainings in the earthquake affected areas, as a result building construction practice has improved a lot. The focus is on the use of horizontal band at plinth, sill, lintel and floor level and vertical reinforcement at corners, junctions of walls and at jambs of opening. The use of corner and through stones is also practiced a lot in new constructions. Stone is dressed as practically possible to produce flat and regular size if it is of relatively bigger stones. In some areas very thin and small round stones are still used. Though the buildings are not fully code compliant, the construction practice has been improved a lot.

Figure 1. Sample Reconstruction of Stone in Mud and Dry Stone Masonry after Gorkha Earthquake

Guideline for low strength masonry in Nepal was developed in Nepal in 1993 but was not followed significantly for the construction of rural dwellings. There is no other code and guidelines that clearly describes the requirements for low strength masonry buildings such as stone in mud and dry stones. Very few requirements and provisions are mentioned in EERI and Indian guideline on this type of a building structure but is certainly inadequate. Chinese code on stone masonry buildings is also cement based. Hence there is no clear guidance for the construction of low strength stone masonry buildings nor much study on such buildings.

2. Experimental Tests on Wall Models

Push-over tests of walls of size 2400 mm x 2100 mm X 400mm with 11 different options were conducted to understand the behavior of various strengthening options before shake table testing of buildings. The tests were performed in displacement control mode with the cyclic lateral –load applied by a hydraulic jack at interval steps of 2mm till the failure of the wall. The failure modes and lateral force-lateral displacement hysteresis diagrams obtained from cyclic tests can be used for the evaluation of the seismic performance of masonry walls. From the force-lateral displacement diagrams, experimental envelopes are derived and the values of the lateral strength and stiffness are calculated. A comparative analysis of the capacity to dissipate energy and ductility for the different types of walls is also carried out. Such mechanical properties of walling system is necessary for simulation of the buildings for seismic evaluation.

The different wall test options and the test results are shown in the table below:

Description			Ultimate lateral displacement mm	Increase ration of the maximum loading	Increase ratio of the Ultimate displacement
Round Stone Wall	Wall #1	Unreinforced	32	-	-
	Wall #2	Wooden Bands & Post + Gabion Wire Meshing	66	48.9%	106.3%
	Wall #3	NF Adhesive + Tarpaulin	68	102.1%	112.5%
	Wall #4	Wooden Post + layer of Gabion Wire at sill and lintel level	48	-14.9%	50.0%
Flat Stone Wall	Wall #5	Unreinforced (NR)	38		
	Wall #6	Built-in wooden bands & post (WB)	57	0	52.6%
	Wall #7	Built-in wooden bandages with gabion wire meshing (WB+GWJ)	134	24.3%	252.6%
	Wall #8	Built-in gabion wire bandage (GWB)	89	18.6%	136.8%
	Wall #9	Wooden bandages with gabion wire meshing and bandage (WB+GWJ+GWB)	238	84.3%	526.3%
	Wall #10	Gabion wire meshing and bandages (GWJ+GWB)	185	118.6%	389.5%
	Wall #11	Horizontally wrapped tarpaulin strips (TS)	100	37.5%	71.4%

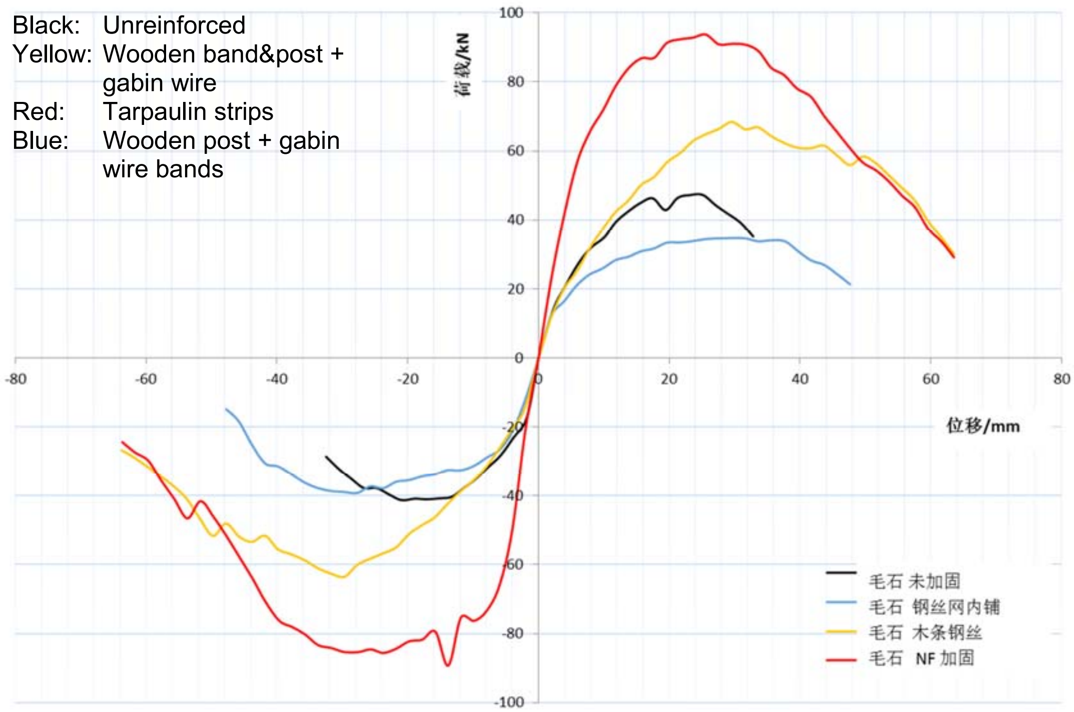


Figure 2. Comparison of loading-displacement curves of round stone wall specimens

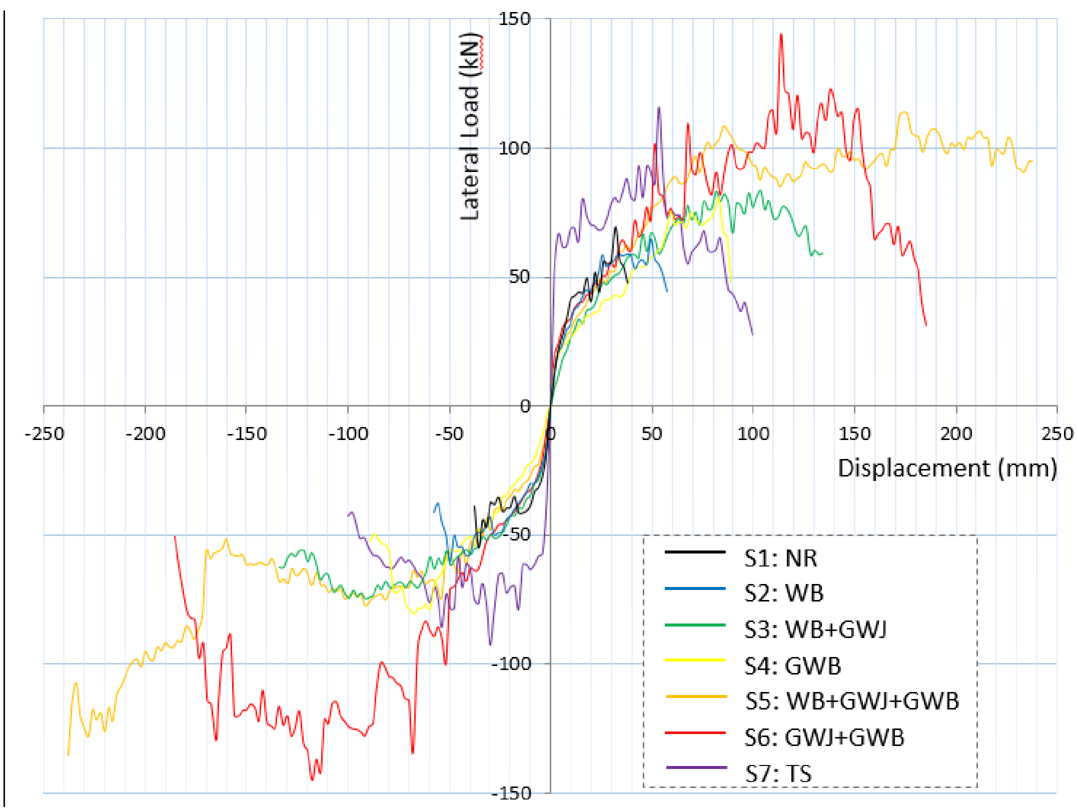


Figure 3. Comparison of loading-displacement curves of flat stone wall specimens

5 SHAKE TABLE TESTS

5.1 Description of Shake Table

The shake table is 4mX4m in plan size. Following table shows the description of Shake table:

Size (Plan)	4mX4m		
Maximum Test Weight (unidirectional)	30 ton (Crane capacity 20 ton)		
Maximum Test Weight (two directional)	30 ton (Crane capacity 20 ton)		
Number of degree of freedom	3 (X-Horiz; Y-Horiz; Rotation)		
Shaking Direction	X-Horz	Y-Horiz	Z-Vert
Max. Acc (at max. loading)	0.8g (test weight of 30t) 1.0g (test weight of 20t)	0.8g (test weight of 30t) 1.0g (test weight of 20t)	
Maximum Velocity	0.8 m/s	0.8 m/s	
Maximum Displacement	250 mm	250 mm	
Existing Frequency	0 ~ 100 Hz	0 ~ 100 Hz	

5.2 Testing of The Building Models

Shake table tests on three types of stone masonry houses were conducted. These tests were completed for two story stone masonry buildings, 3/5 scale in shake table of Kunming University of Science and Technology with the input data of El-Centro earthquake at incremental loading. The size of the room for full scale is 4.5 m X 4.5 m and wall thickness 450 mm. The building models were tested using well-dressed stones and proper through and corner stitches. The three building models are 1) Unreinforced stone masonry (Model 1) 2) Stone masonry with wooden bands and posts as per NBC:203 (Model 2) and 3) Stone masonry with wooden bands and posts plus gabion wire meshing (Model 3). The first shake table test of Model 1 was conducted on 19th September 2016 in presence of the representatives from Beijing Normal University (BNU), Kunming University of Science and Technology (KUST) and National Society for Earthquake Technology-Nepal (NSET). The second and the third building models Model 2 and Model 3 were tested on 21 st and 22nd January 2017 at KUST in presence of the large group of representatives from the Government of Nepal, faculty members from the universities of Nepal, Bangladesh and China, people from fields and project team members.

The outcome of the experimental tests are 1) The unreinforced stone masonry building showed life safety at 0.22g and partial collapse of the structure at 0.4g. 2) The same building with timber as earthquake resistant elements as per Nepal National Building Code NBC: 203 also performed well up to 0.62g allowing the structure to be stable. But there was no Life Safety in the building due to local failure of stone walls in between horizontal and vertical bands. Stones tend to push out at low intensity of shaking and failed at 0.51 g. 3). The building model with wooden bands and gabion wire performed well and met the expectation preventing local failure of walls. The building was excited up to 1.0 g in X-direction and 0.92 g in Y direction with Life Safety in the building. The effect of gabion wire on both the sides of the wall, the ability to prevent the local failure, was well observed in the test.

The summary of the failure modes of the buildings is given in Table below.

Building Model	Failure Mode at the Expected level of Shaking
Unreinforced (Model 1)	Out-of plane failure in upper story
Wooden Bands and Posts as per NBC 203 (Model 2)	Local failure of walls in between the bands
Wooden Bands and Posts and gabion wire meshing (Model 3)	Life Safety with sufficient ductility

	
Damage to **Unreinforced Building** at 0.31g	Damage to **Unreinforced Masonry** at 0.40g

	
Damage to **the Building with Wooden Bands and Posts** at 0.51g	Damage to the **Building with Wooden Bands and Posts and gabion wire meshing** at 0.62g

6 REMAINING PROJECT TASKS

The validation with experimental tests will be compared with analysis results of computer models that NSET will continue to work. The next step is to disseminate the lessons learnt from experimental tests to the technical and non-technical people in field for implementation. This will be done by developing a guideline and easy to read handbooks on the construction of earthquake resistant stone masonry buildings in local language. More specifically, typical design with all necessary details on the recommended improved technology will be prepared and submitted to MoUD, NRA for inclusion in the design catalogue. This is planned in February and March 2017. At the same time it is very important to mainstream the technology of building construction in government published documents. The right technology should be propagated at the local level with proper institutionalization. For this the advocacy to key government organizations will be continued to incorporate the improved technology on low strength two story stone masonry buildings which proved satisfactory in shake table test. The final step is on-site demonstration and transfer of appropriate technology at local level. The site will be selected for demonstration.

7 SUMMARY AND CONCLUSIONS

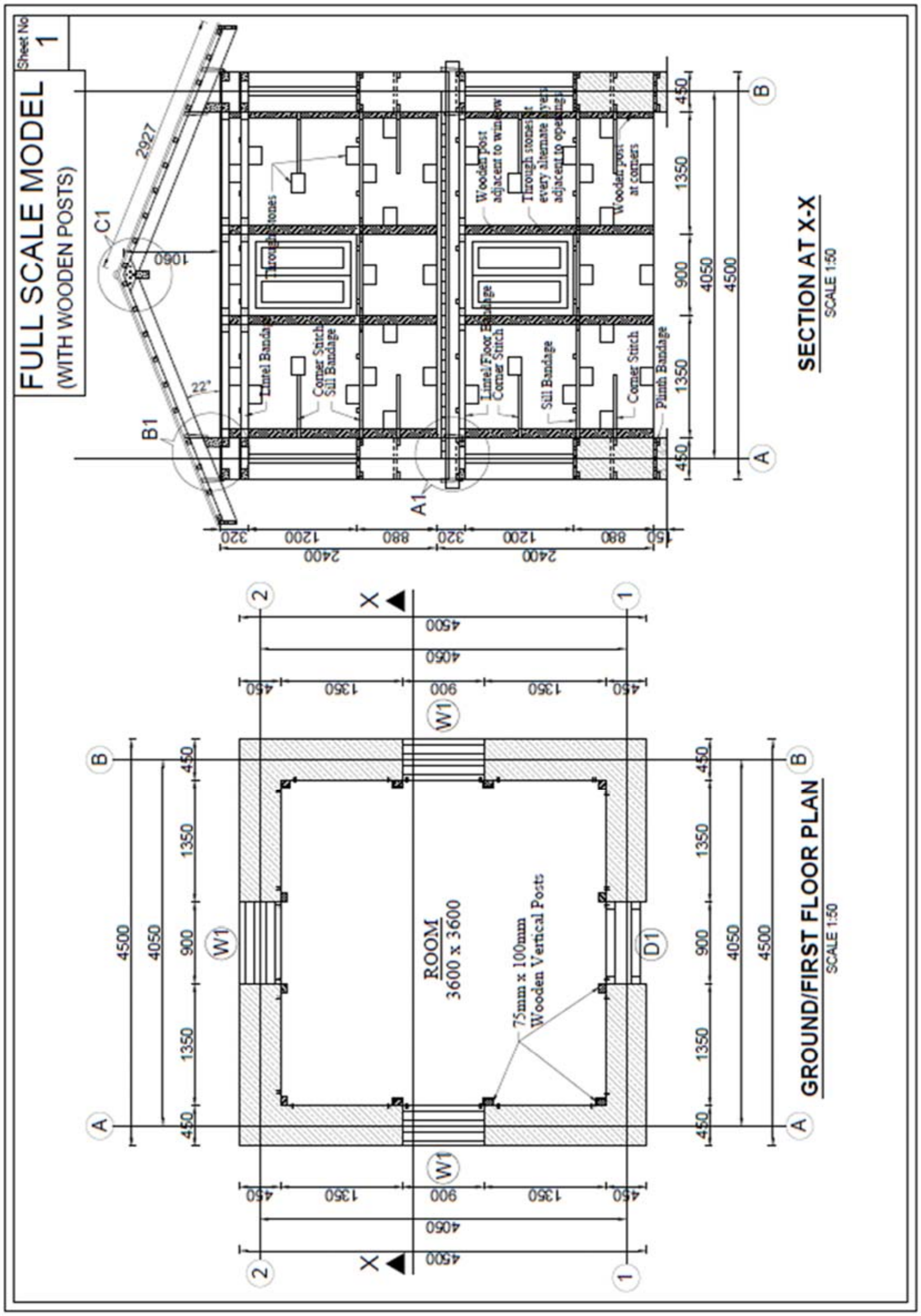
There is no available guidelines worldwide on low strength stone masonry buildings for high seismic zones except the Nepal National Building Code NBC: 203. Very few requirements and provisions are mentioned in EERI and Indian guideline on this type of a building structure but is largely inadequate. The NBC code was developed 24 years ago with the recent update of the code in 2015 and is based on analytical and conceptual solutions.

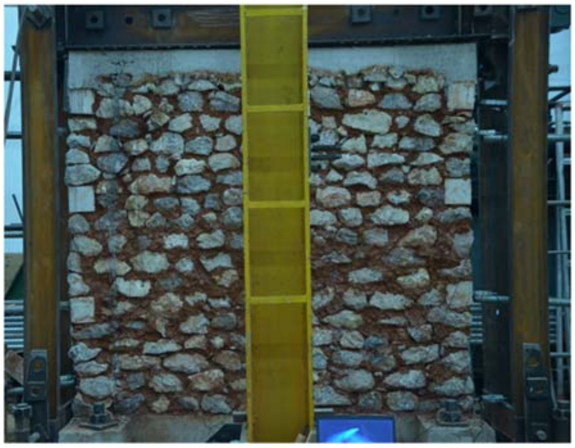
The design ground acceleration as per Nepal National Building Code for 500 year return period vary between 0.3 g and 0.42 g for medium soil condition depending on the location of the building. Since Nepal lies in a high seismic zone, it is expected that the peak ground acceleration at a particular site can exceed the design ground acceleration. So the following conclusions can be drawn from the experimental tests for the expected level of earthquake shaking.

1. Unreinforced stone masonry building (Model 1) cannot ensure Life Safety in the building.

2. Stone masonry building with wooden bands and posts as per NBC: 203 (Model 2) also cannot ensure Life Safety in the building due to the local failure of stones in between the bands. The more irregular and round the stones are used, more is the risk to the building occupants due to collapse of stone walls in between the bands.
3. Stone masonry with wooden bands and posts as per NBC: 203 and gabion wire meshing (Model 3) can ensure Life Safety in the building with sufficient ductility.
4. The tested building models were well built with properly dressed, regular and flat stones, through and corner stones and stitches, horizontal and vertical timber bands as per building code NBC 203. But in actual practice at site, the construction of buildings is not as good as the tested models. In case of random rubble, irregular and slightly round stones in building, the test indicates more easy local failure of stones even at low intensities of earthquake shaking. Since there is practically no bonding between stone elements, it is always a good practice to use gabion wires on either sides of the walls to prevent the local failure of stone walls. Such gabion wires should be close enough to confine the materials inside and well connected with the wall. Hence the use of earthquake resistant elements as given in NBC 203 with addition of gabion wire on either sides of the stone walls meet the code level expectation of earthquake safety in stone masonry buildings in Nepal and other Himalayan zones and is encouraged to use in the construction of such buildings including buildings for reconstruction.

ANNEX 1. FULL SCALE BUILDING DRAWING FOR MODEL 2





Wall # 1 Random Rubble Stone Wall before Testing

Movement of the wall at ultimate displacement of 32 mm

Wall #1 Ultimate Crack Pattern of Random Rubble Stone Wall

Wall # 2 Random Rubble with Wooden Band and post and Gabion Wire Meshing

Movement of the wall at ultimate displacement of 66 mm

Wall # 2: Stretching of Gabion Wire and Bulging out of Stone Wall

Wall #6: Unreinforced Flat Stone

Movement of the wall at ultimate displacement of 48 mm with local failure of random rubble stone wall

Wall #6: Ultimate failure of Unreinforced Flat Stone

ANNEX 2. SOME PHOTOGRAPHS OF WALL TESTS

ANNEX 3. BRIEF PROJECT DESCRIPTION

A) Key Project Information

Tittle:	**Development, testing, demonstration and training of better-built procedures and retrofit techniques for non-engineered housing in urban and rural areas of the Himalayan belt.**	
Duration:	July 2016 to September 2017.	
Project Partners:		
	Funding	DFID of UK
	Partnering institutions	Beijing Normal University (VNU) Tsinghua University (THU) Kunming University of Science and Technology (KUST) National Society for Earthquake Technology-Nepal (NSET) Institute of Engineering (IOE)
MOU Signed Date:	July 2016	
Milestones: • On-site demonstration on how to do it: learning, • Development of training materials for masons and engineers, • Training of trainers and • Improvement of existing building codes and retrofit guideline in Nepal		
Advisory Committee:	Prof. Dina D'Ayala (University College London) Dr. Ramesh Guragain (NSET) Prof. Khondoker Mokaddem Hossain (University of Dhaka) Prof. Bing Li (NHRC) Prof. Ning Li (BNU)	
Key Project Staffs:	Prof. Ming Wang (BNU) Hima Shrestha (NSET) Kuber Bogati (NSET) Prof. Gokarna Motra (IOE)	

B) Schedules of Shake Table Tests

Building Model and Test Date	Name of the Participants	Origination
Unreinforced (Model 1) Test date : September 19, 2016	Hima Gurubacharya Ranjan Dhungel	NSET-Nepal
	Ming Wang Kai Liu	BNU- China
	Zhilang Niu Wen Pan Xiaodong Yang	KUST-China

Wooden Bands and Posts as per NBC 203 (Model 2)	Su Zhang	(DFID-China)
Test date : January 21, 2017	Xiao Zheng	(UNDP-China)
	Yi Yuan Hongjian Zhou	(National Disaster Reduction Centre of China)
Wooden Bands and Posts and gabion wire meshing (Model 3)	Yaqiao Wu	ICCR-DRR-China
Test date : January 22, 2017	Zhilang Niu Wen Pan Xiaodong Yang	KUST-China
	Ming Wang Kai Liu	BNU- China
	Hari Ram Parajuli	NRA-Nepal
	Gokarna Motra	IOE-Nepapl
	Suman Salike	Ministry of Urban Development, Nepal
	Manoj Nakarmi	DUDBC-Nepal
	Parikshit Parik Kadariya	CLPIU (MoUD),Nepal
	Manohar Rajbhandari	Consultant
	Dr. Ramesh Guragain Surya Narayan Shrestha Hima Gurubacharya Rajani Prajapati Kuber Bogati	NSET-Nepal
	Jyoti Mani Bhattarai Achyut Paudel Ayush Baskota Pramod Khatiwada	Reconstruction Program, NSET-Nepal

C) Shake Table Test Cases

Case No	Case Name	Waves	Direction	PTA(g)
1	W1	1st White noise	XY	0.05
2	E-Xy- 0.1	Elcentro (NS-EW)	X	0.1
			Y	0.085
3	E-YX- 0.1	Elcentro (NS-EW)	X	0.085
			Y	0.1
4	W2	2nd White noise	XY	0.05
5	E-Xy- 0.15	Elcentro (NS-EW)	X	0.15

				Y	0.1275
6	E-YX-	0.15	Elcentro (NS-EW)	X	0.1275
				Y	0.15
7	W3		3rd White noise	XY	0.05
8	E-Xy-	0.22	Elcentro (NS-EW)	X	0.22
				Y	0.187
9	E-YX-	0.22	Elcentro (NS-EW)	X	0.187
				Y	0.22
10	W4		4th White noise	XY	0.05
11	E-Xy-	0.31	Elcentro (NS-EW)	X	0.31
				Y	0.2635
12	E-YX-	0.31	Elcentro (NS-EW)	X	0.2635
				Y	0.31
13	W5		5th White noise	XY	0.05
14	E-Xy-	0.4	Elcentro (NS-EW)	X	0.4
				Y	0.34
15	E-YX-	0.4	Elcentro (NS-EW)	X	0.34
				Y	0.4
16	W6		6th White noise	XY	0.05
17	E-Xy-	0.51	Elcentro (NS-EW)	X	0.51
				Y	0.4335
18	E-YX-	0.51	Elcentro (NS-EW)	X	0.4335
				Y	0.51
19	W5		7th White noise	XY	0.05
20	E-Xy-	0.62	Elcentro (NS-EW)	X	0.62
				Y	0.527
21	E-YX-	0.62	Elcentro (NS-EW)	X	0.527
				Y	0.62
22	W1		8th White noise	XY	0.05
and So on up to 1.0g					

D) Schedules of Remaining Activities

On-site demonstration on how to do it: learning	April 2017~ May 2017
Development of training materials for masons and engineers	March 2017~ June 2017
Training of trainers	July 2017~ Aug 2017

Improvement of existing building codes and retrofit guideline in Nepal	Aug 2017~ Sept 2017